Individual Progress Report

ECE 445 Design Document - Spring 2026

Project #40

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Introduction:

This project, Bilateral Earlobe Pulse Timing Measurement Device, focuses on developing a synchronized physiological sensing platform capable of measuring pulse arrival time (PAT) differences between the left and right earlobes. The system integrates three main subsystems: an ECG analog front-end, dual PPG sensing modules, and a centralized data acquisition and control unit. The ECG subsystem provides a cardiac timing reference through R-peak detection, while the dual PPG subsystems measure blood volume changes at both earlobes. These signals are synchronized through a shared clock and processed by a microcontroller to enable precise bilateral pulse timing comparison .

My primary contribution to this project lies in the hardware implementation of the dual PPG subsystem and its integration with the overall system. Specifically, I was responsible for designing the printed circuit boards (PCBs) for both the ECG board and the PPG boards. The PPG boards interface directly with the optical sensing components and communicate with the central microcontroller subsystem, forming a critical link between physiological signal acquisition and digital processing. This work closely interacts with my teammates' contributions in firmware development, system synchronization, and mobile application integration. For example, the effectiveness of the shared timing architecture and Bluetooth data transmission depends heavily on the correctness of the PCB design, signal routing, and hardware-level reliability of the sensing boards.

So far, I have completed the schematic design, PCB layout, and physical assembly (soldering) of both the ECG and PPG boards. Additionally, I have prepared and flashed the necessary firmware onto a development kit (DK) board to validate basic functionality. Recently, I successfully soldered the PPG board and verified that the program can be uploaded and executed on the DK platform. However, flashing the same program directly onto my custom-designed PCB has not yet been successful, indicating potential issues in hardware connections, bootloading configuration, or debugging interfaces.

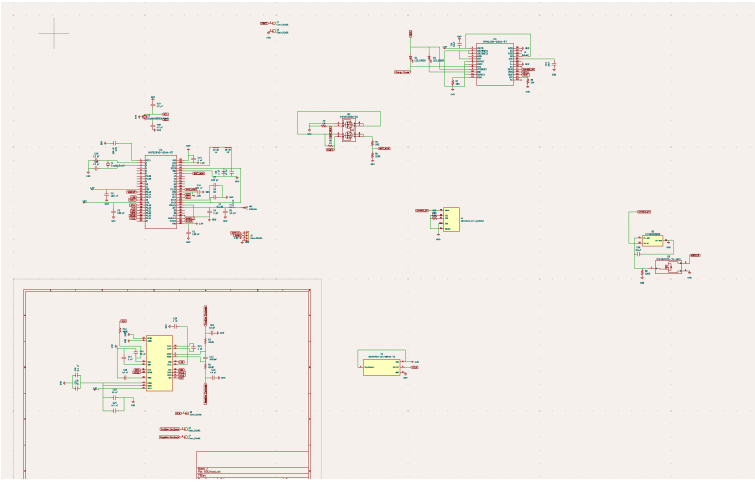
Currently, I am focused on debugging the custom PCB to resolve programming and communication issues. This involves verifying power integrity, checking signal continuity, and ensuring proper connections for programming interfaces such as SWD. In parallel, I am beginning development of a Python-based application that will connect to the system via Bluetooth Low Energy (BLE) to visualize real-time PPG waveforms. This software component is essential for the upcoming progress demo, where we aim to demonstrate successful acquisition and display of physiological signals from the PPG subsystem.

Looking forward, my remaining tasks include fully debugging and validating the PPG board, ensuring reliable firmware deployment on the custom PCB, and completing the BLE-based data visualization pipeline. These efforts will directly support system integration and help demonstrate synchronized physiological sensing as required by the project objectives.

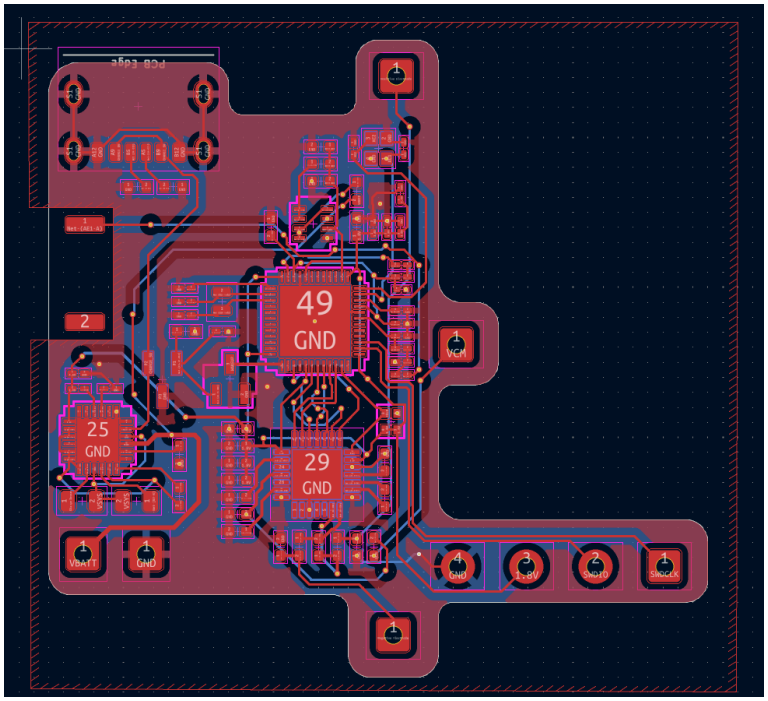
Design:

1. PCB Design Overview

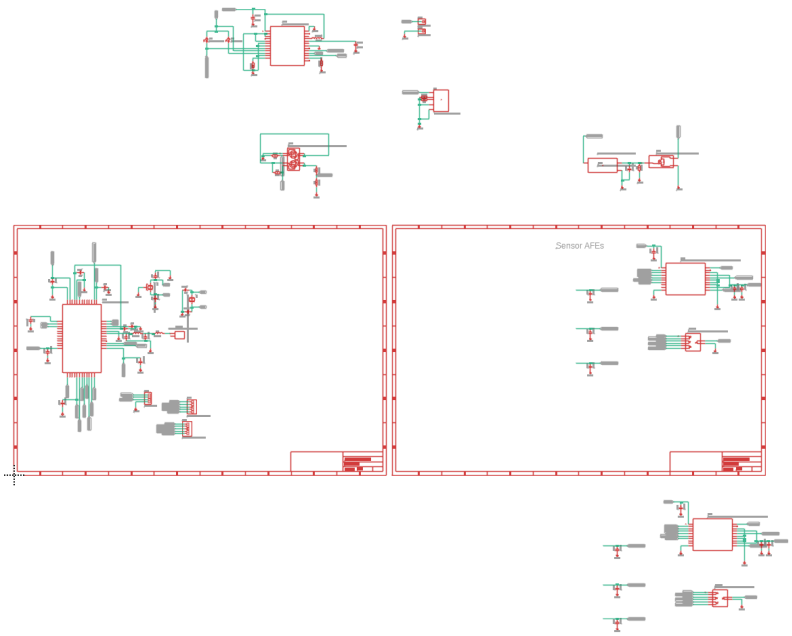
I designed two main custom boards: the ECG board and the PPG board. Both boards are centered around integrating analog front-end (AFE) sensors with the microcontroller unit (MCU) through SPI communication and ensuring signal integrity for low-noise physiological measurements.



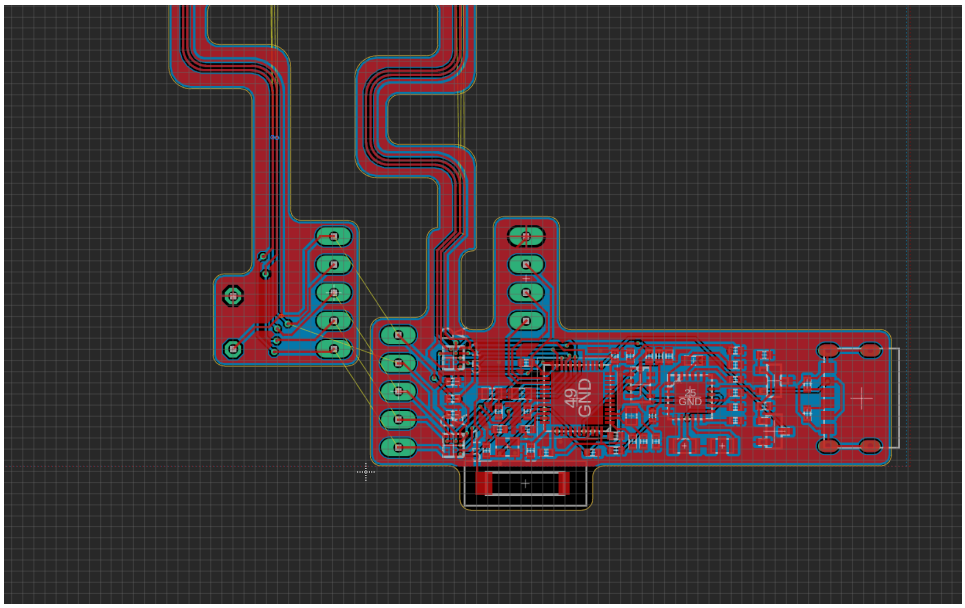
ECG PCB Schematic



ECG PCB Layout / Routing



PPG PCB Schematic



PPG PCB Layout / Routing

The PCB design process included schematic capture, component selection, footprint verification, and multi-layer routing. Special attention was given to separating analog and digital grounds, minimizing noise coupling, and ensuring stable power delivery to sensitive analog components.

2. PPG Board Design (Primary Contribution)

The PPG board is built around the **MAX86141 optical AFE** and the **SFH-7050A optical sensor**, which together enable photoplethysmography signal acquisition.

Circuit Design

The MAX86141 integrates LED drivers, photodiode inputs, and ADC functionality, which simplifies external circuitry requirements. However, several important design considerations were implemented:

- **LED Drive Configuration**:
  The LED current is controlled via internal registers. Based on datasheet recommendations and safety constraints, the drive current is limited to avoid excessive heating on the earlobe.
- **Photodiode Signal Acquisition**:
  The reflected light signal is converted into a digital value through the internal ADC. No external amplification is required, but proper decoupling and grounding are critical.
- **Power Filtering**:
  Local decoupling capacitors (0.1 μF and 1 μF) were placed close to the power pins of the MAX86141 to reduce supply noise.
- **SPI Communication Interface**:
  The PPG board communicates with the MCU via SPI. Signal lines (MOSI, MISO, SCLK, CS) were routed with controlled impedance and minimal length mismatch to ensure reliable high-speed communication.

PCB Layout Considerations

- The analog front-end region was physically isolated from digital traces.
- Ground planes were used to reduce electromagnetic interference (EMI).
- The optical sensor placement was optimized to align with the earlobe clip for consistent signal acquisition.
- Short trace lengths were used for sensitive analog paths.

Interface with System

The PPG board connects to the central MCU (nRF52840) and relies on:

- SPI bus for data transfer
- Interrupt line (INT) for data-ready signaling
- Shared power rail (3.3V)
- Common system clock for synchronization

This ensures that both PPG channels can be sampled simultaneously and aligned with the ECG signal for accurate pulse timing analysis.

3. ECG Board Design

The ECG board is responsible for acquiring low-amplitude biopotential signals from the human body and conditioning them for digital conversion.

Circuit Design

The ECG front-end is based on the **MAX30003 biopotential AFE**, which provides:

- High input impedance for electrode signals
- Built-in amplification and filtering
- R-peak detection with interrupt output

Key design elements include:

- **Input Protection and Safety**:
  Series resistors were used to limit current and ensure patient safety.
- **Filtering**:
  Internal and external filtering components help remove baseline drift and high-frequency noise.
- **Clock Source**:
  A dedicated low-frequency crystal oscillator (32.768 kHz) was used to provide accurate timing for ECG signal acquisition.

PCB Layout Considerations

- Differential signal routing for electrode inputs to reduce common-mode noise
- Placement of analog components close to the AFE
- Careful grounding strategy to avoid noise injection from digital circuits

Interface with System

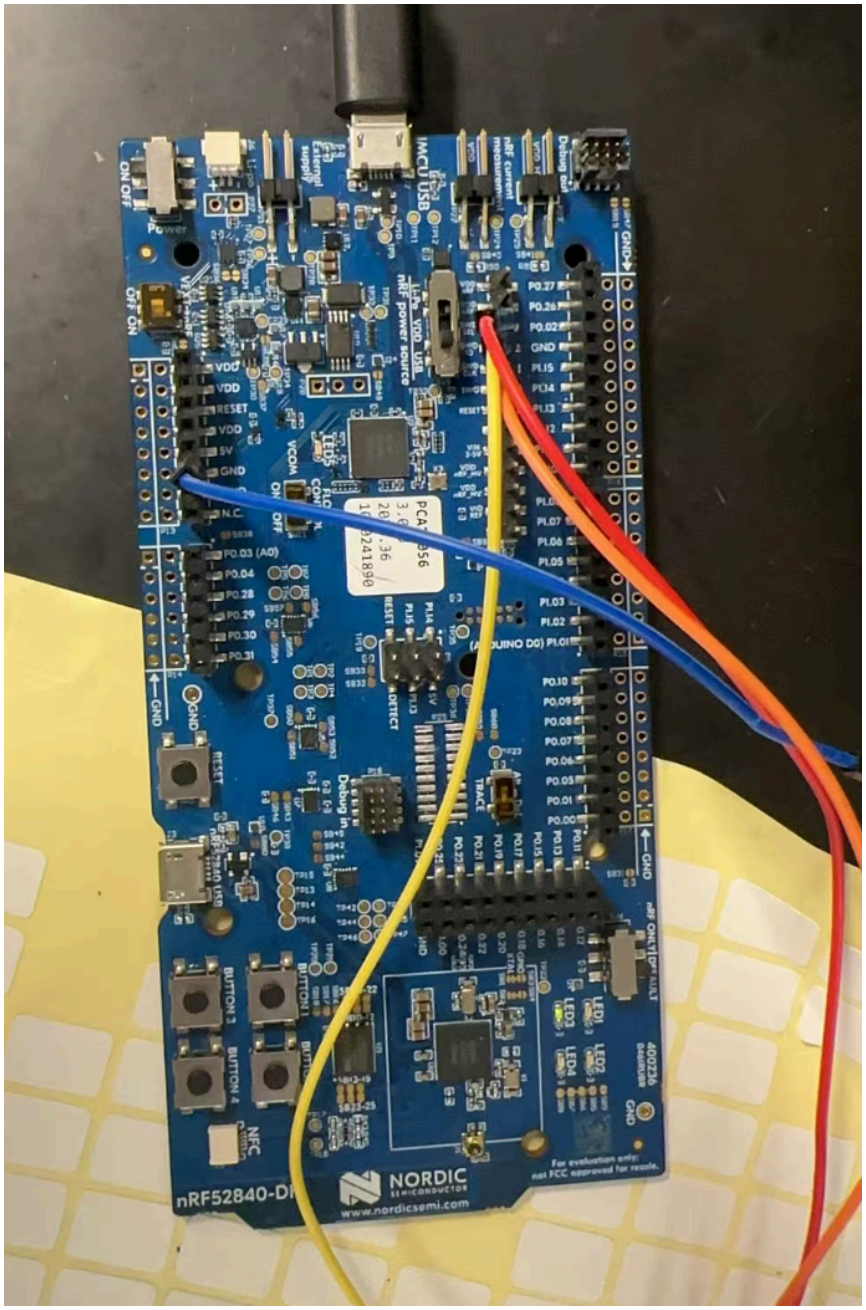
The ECG board interfaces with the MCU through:

- SPI communication for data acquisition
- Interrupt signal for R-peak detection
- Shared timing reference for synchronization with PPG channels

This ECG signal serves as the reference point for pulse transit time (PTT) calculations.

4. Firmware Flashing and Hardware Interface

In addition to hardware design, I worked on preparing the firmware flashing setup for the custom PCBs.



Firmware flashing setup / DK connection

The system uses the nRF52840 MCU, which is programmed via the SWD interface. I verified that:

- The firmware can be successfully compiled and flashed onto a development kit (DK)
- SPI communication and basic peripheral initialization work correctly on the DK

However, flashing the same firmware onto the custom PCB has not yet been successful. This suggests potential issues such as:

- Incorrect SWD pin routing
- Power supply instability
- Missing or incorrect pull-up/down resistors
- Boot configuration or clock initialization issues

Debugging this interface is currently ongoing.

5. Design Integration with Other Subsystems

My hardware design directly supports and interacts with other team components:

- **Firmware Team**:
  My PCB design must correctly expose SPI, interrupt, and power interfaces to allow firmware to configure and read from the sensors.
- **Synchronization Architecture**:
  The shared clock and interrupt-driven design enable precise alignment of ECG and PPG signals.
- **Bluetooth Data Transmission**:
  Clean and reliable signal acquisition from my boards is essential for accurate data packaging and transmission via BLE.
- **Software Visualization (Upcoming Work)**:
  The data generated by my hardware will be used by a Python-based BLE application to display real-time PPG waveforms.

Verification:

Verification of my work primarily focused on ensuring the correctness of the PCB assembly, validating power integrity, and confirming that the hardware is capable of supporting firmware flashing and sensor operation. Since my contributions are centered on hardware implementation (PCB design, soldering, and initial bring-up), the testing process emphasized electrical validation, connectivity checks, and functional readiness for system integration.

1. Power Integrity Testing

One of the most critical requirements for successful operation of the PPG and ECG boards is stable and correct power delivery, particularly for the microcontroller and analog front-end components.

Test Procedure

- A digital multimeter (DMM) was used to probe key power rails across the PCB.
- Measurements were taken at:
  - The output of the voltage regulator

    - The VDD pins of the MCU and sensor ICs
    - Decoupling capacitor terminals near critical components
- The expected voltage for the digital supply rail (VDD) was **1.8V**, which is required for proper operation of the MCU and successful firmware flashing.

Results

- The measured VDD voltage was consistently around **1.8V ± small tolerance**, indicating that the voltage regulator and power distribution network are functioning correctly.
- No significant voltage drops were observed across traces, suggesting good solder joints and adequate power routing.

Significance

Ensuring a stable 1.8V supply is essential because:

- Undervoltage would prevent the MCU from entering programming mode.
- Overvoltage could damage sensitive components.
- Stable power confirms that the board is electrically viable for further debugging and firmware deployment.

2. Continuity and Connectivity Testing

To verify that the PCB was assembled correctly and that there were no open circuits or unintended shorts, extensive continuity testing was performed.

Test Procedure

- The DMM was set to continuity mode.
- Critical signal paths were tested, including:
    - Power and ground connections
    - SPI communication lines (MOSI, MISO, SCLK, CS)
    - SWD programming interface (SWDIO, SWCLK, GND, VDD)
    - Interrupt lines from sensors to MCU
- Each trace was probed from the source pin (e.g., MCU) to the destination pin (e.g., sensor IC).

Results

- All major signal paths showed proper continuity (audible beep / near-zero resistance).
- No unintended shorts were detected between adjacent pins or between power and ground.
- This confirms that the PCB routing and soldering are electrically consistent with the schematic.

Significance

These tests validate:

- Correct physical assembly of the PCB
- Proper soldering of fine-pitch components
- Reliability of communication interfaces required for system operation

3. Firmware Flashing Verification (Partial)

A key functional requirement is the ability to flash firmware onto the custom PCB.

Test Procedure

- Firmware was compiled and successfully flashed onto a development kit (DK) using the same MCU (nRF52840).
- The same flashing setup (SWD interface) was then applied to the custom PCB.
- Connections between the debugger and PCB were verified using continuity testing prior to flashing attempts.

Results

- Firmware flashing was **successful on the DK board**, confirming that the software and toolchain are correctly configured.
- However, flashing to the custom PCB has **not yet been successful**.

Analysis

Given that:

- Power rails are correct (verified via DMM)
- SWD connections show continuity

The remaining possible issues include:

- Signal integrity issues on SWD lines
- Incorrect boot configuration or missing pull resistors
- Clock initialization problems
- Subtle soldering defects
- This indicates that while the hardware is partially validated, further debugging is required before full functionality is achieved.

4. PPG Board Functional Bring-Up (Initial Testing)

After soldering the PPG board, initial functional testing was conducted.

Test Procedure

- The firmware was loaded onto the DK board.

- The DK was connected to the PPG board to test sensor communication.
- Basic SPI communication and initialization routines were executed.

Results

- The firmware successfully ran on the DK and interfaced with the system.
- This confirms that:
    - The software stack is functional
    - The system architecture is valid

However:

- Full standalone operation on the custom PCB is still pending successful flashing.

5. Tests Yet to Be Performed

Several important verification steps remain to fully validate my subsystem:

(1) Successful Firmware Flashing on Custom PCB

- Confirm that the MCU can be programmed via SWD on the custom board.
- This is a critical milestone for full system operation.

(2) Sensor Data Validation

- Verify that the PPG sensor produces valid waveform data.
- Measure signal amplitude and noise levels.
- Confirm that the waveform reflects physiological pulse signals.

(3) BLE Communication Testing

- Establish Bluetooth Low Energy connection with an external device.
- Validate real-time data transmission.

(4) End-to-End System Testing

- Synchronize ECG and PPG data streams.
- Verify timing alignment and data integrity.

(5) Long-Term Stability Testing

- Ensure consistent operation over extended periods.
- Monitor for thermal issues or signal drift.

Conclusion:

To complete my responsibilities, my immediate focus is debugging the custom PCB to enable successful firmware flashing. Although power and continuity have been verified, I will next check SWD signals, clock functionality, and possible soldering issues. After resolving this, I plan to validate SPI communication with the PPG sensor and obtain real waveform data.

Before the progress demo, I will develop a Python-based BLE application to display real-time PPG signals, demonstrating that the hardware can successfully acquire and transmit physiological data. In the final stage, I will support full system integration and ensure stable operation of both PPG channels alongside the ECG subsystem.

From an ethical perspective, this project follows principles from the IEEE and ACM Codes of Ethics. We ensure honesty by clearly stating that this is a research prototype, not a medical device. Safety is addressed through low-voltage design and current-limiting measures. Additionally, since physiological data is involved, we consider privacy by limiting data use and ensuring responsible handling. Overall, the project emphasizes both technical reliability and responsible engineering practice.

Citations:

1. IEEE. *IEEE Code of Ethics*. IEEE, 2020. https://www.ieee.org/about/corporate/governance/p7-8.html
2. ACM. *ACM Code of Ethics and Professional Conduct*. Association for Computing Machinery, 2018. https://www.acm.org/code-of-ethics
3. Analog Devices. *MAX86140/MAX86141 Datasheet: Optical Data Acquisition System*. 2023.
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6. ams-OSRAM. *SFH 7050A Optical Sensor Datasheet*. 2023.
7. MAX30003. *MAX30003 Ultra-Low Power, Single-Channel Integrated Biopotential AFE Datasheet*. Analog Devices, Rev. 2, 2024.